VISUALISING THE GREATEST OF FOOTBALL

An Analysis of the Premier League 1993 to 2023

THE PREMIER LEAGUE

- Highest level of the English football league system
- Founded in 1992
- Contested by 20 clubs over a full year
- System of promotion and relegation
- Most-watched and most popular sports league in the entire world

A BRIEF OVERVIEW OF THE DATASET

Field	Description	Data Type
Season_End_Year	The year the season ends	Numerical
Wk	The week of the season	Numerical
Date	The match date	Temporal
Home	Home team	Categorical
HomeGoals	Goals scored by the home team	Numerical
AwayGoals	Goals scored by the away team	Numerical
Away	Away team	Categorical
FTR	Full-time result	Categorical

A BRIEF OVERVIEW OF THE DATASET

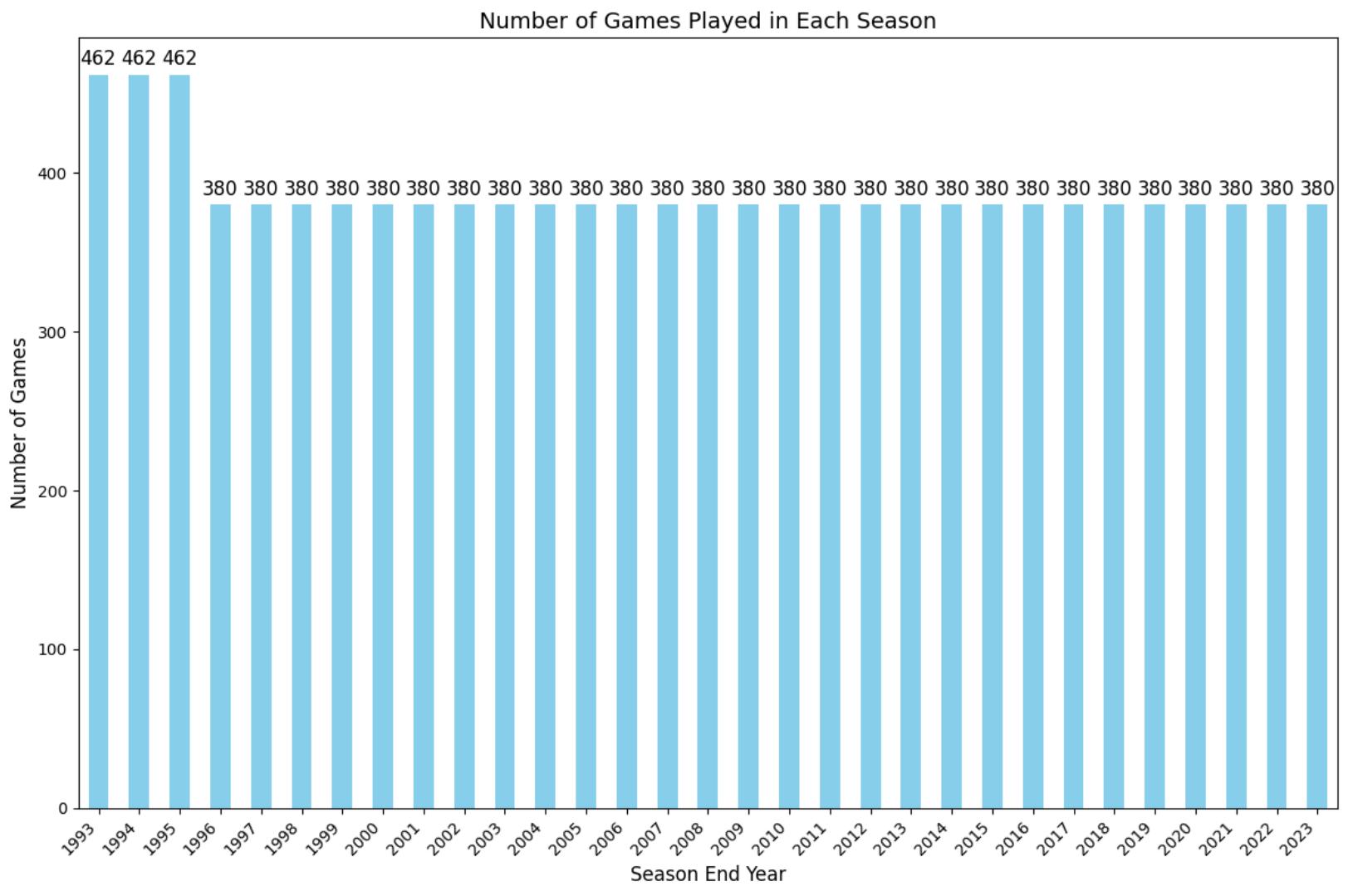
DATASET SHAPE — (12026, 8)

FIRST TEN ROWS

	Season_End_Year	Wk	Date	Home	HomeGoals	AwayGoals	Away	FTR
0	1993	1	1992-08-15	Coventry City	2	1	Middlesbrough	H
1	1993	1	1992-08-15	Leeds United	2	1	Wimbledon	H
2	1993	1	1992-08-15	Sheffield Utd	2	1	Manchester Utd	H
3	1993	1	1992-08-15	Crystal Palace	3	3	Blackburn	D
4	1993	1	1992-08-15	Arsenal	2	4	Norwich City	A
5	1993	1	1992-08-15	Ipswich Town	1	1	Aston Villa	D
6	1993	1	1992-08-15	Everton	1	1	Sheffield Weds	D
7	1993	1	1992-08-15	Southampton	0	0	Tottenham	D
8	1993	1	1992-08-15	Chelsea	1	1	Oldham Athletic	D
9	1993	1	1992-08-16	Nott'ham Forest	1	0	Liverpool	H

Covers all the matches held from the 1992-1993 season to the 2022-2023 season, spanning 31 seasons in total

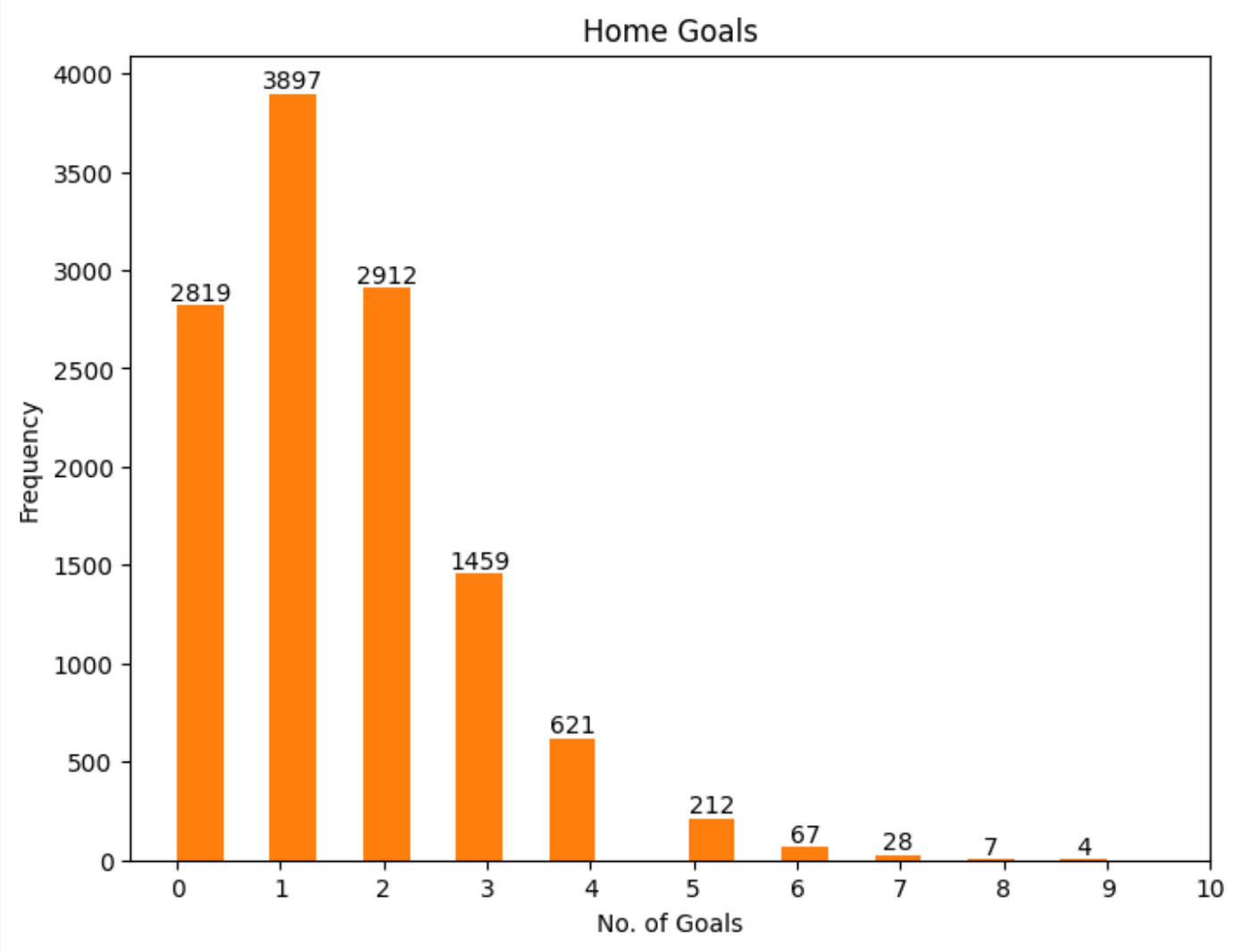
INITIAL VISUALISATIONS



462 games in the first 3 seasons – 22 teams

380 games ever since – 20 teams per season

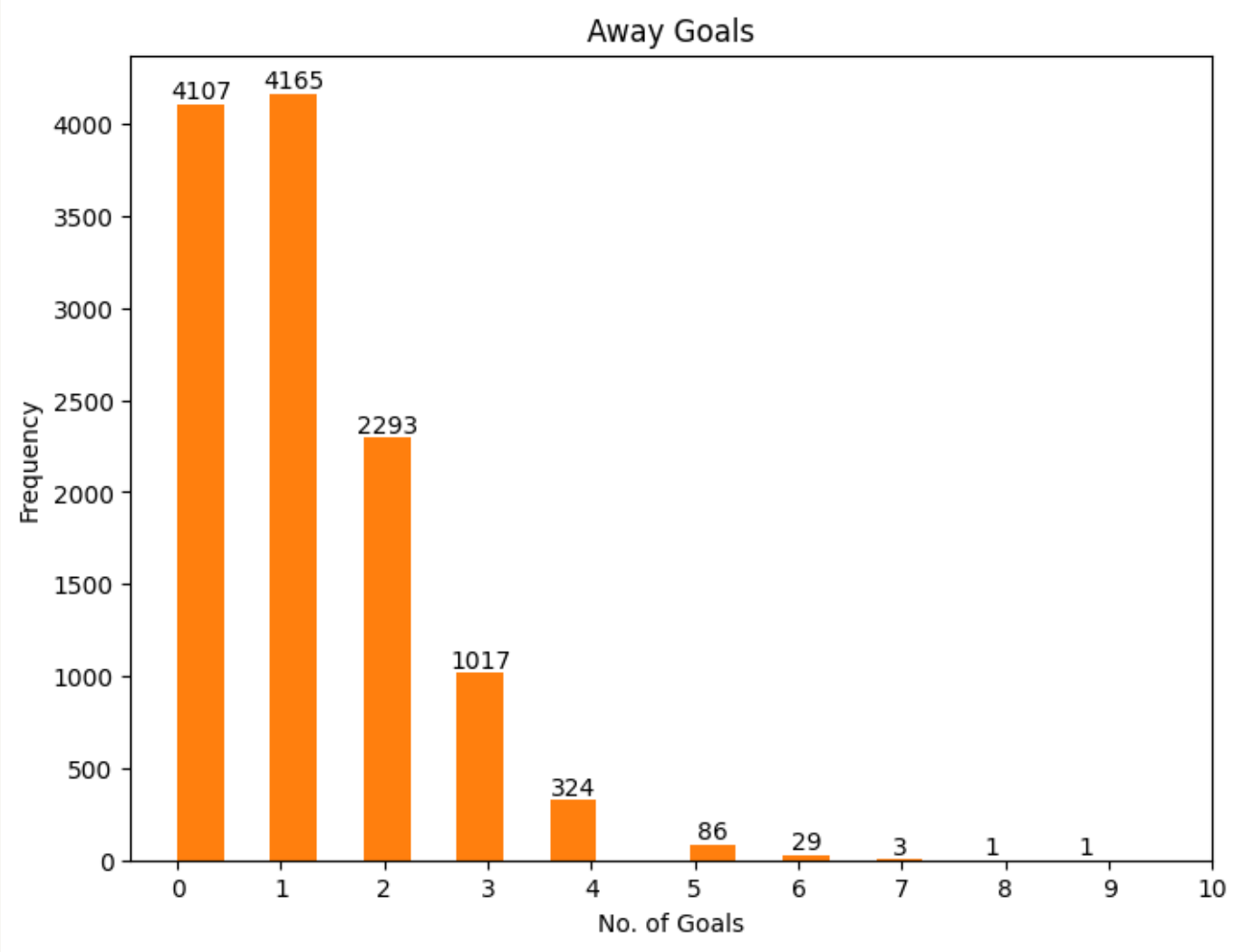
INITIAL VISUALISATIONS



Home teams are generally assumed to have an advantage in any competition

1 to 2 goals are scored by home teams on average every game – 1.52, to be precise

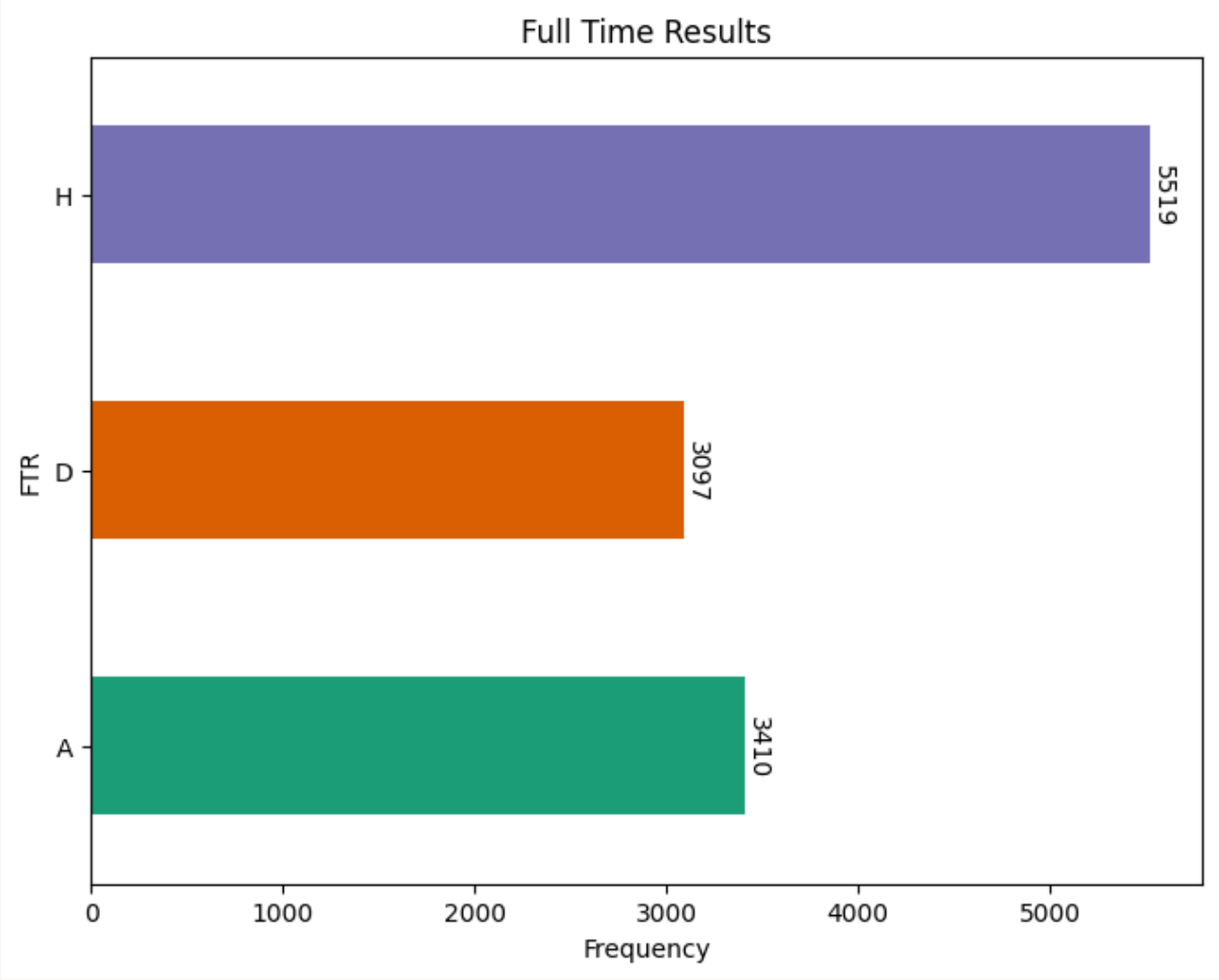
INITIAL VISUALISATIONS



Average number of goals by away teams is 1.14, less than that of the home team

Doesn't mean the away team always loses, but it does showcase that they always do find it more challenging to win

INITIAL VISUALISATIONS



As expected, the home team has won more matches when compared to the away team.

Percentage of matches in which the home team has remained unbeaten (wins and draws) is 71%.

Can be visualised in a pie chart too, but could lead to confusion without data labels.

CONVERSION TO A POINTS TABLE

Team	Played	Won	Drawn	Lost	GF	GA	GD	Points	Season_End_Year	Position
Liverpool	38	32	3	3	85	33	52	99	2020	1
Manchester City	38	26	3	9	102	35	67	81	2020	2
Manchester Utd	38	18	12	8	66	36	30	66	2020	3
Chelsea	38	20	6	12	69	54	15	66	2020	4
Leicester City	38	18	8	12	67	41	26	62	2020	5
Tottenham	38	16	11	11	61	47	14	59	2020	6
Wolves	38	15	14	9	51	40	11	59	2020	7
Arsenal	38	14	14	10	56	48	8	56	2020	8
Sheffield Utd	38	14	12	12	39	39	0	54	2020	9
Burnley	38	15	9	14	43	50	-7	54	2020	10
Southampton	38	15	7	16	51	60	-9	52	2020	11
Everton	38	13	10	15	44	56	-12	49	2020	12
Newcastle Utd	38	11	11	16	38	58	-20	44	2020	13
Crystal Palace	38	11	10	17	31	50	-19	43	2020	14
Brighton	38	9	14	15	39	54	-15	41	2020	15
West Ham	38	10	9	19	49	62	-13	39	2020	16
Aston Villa	38	9	8	21	41	67	-26	35	2020	17
Bournemouth	38	9	7	22	40	65	-25	34	2020	18
Watford	38	8	10	20	36	64	-28	34	2020	19
Norwich City	38	5	6	27	26	75	-49	21	2020	20

Reshaping the existing data; generating points tables for each season

Conversion using a simple Python script, provides a new visualisation

Can further be used to generate more visualisations and insights

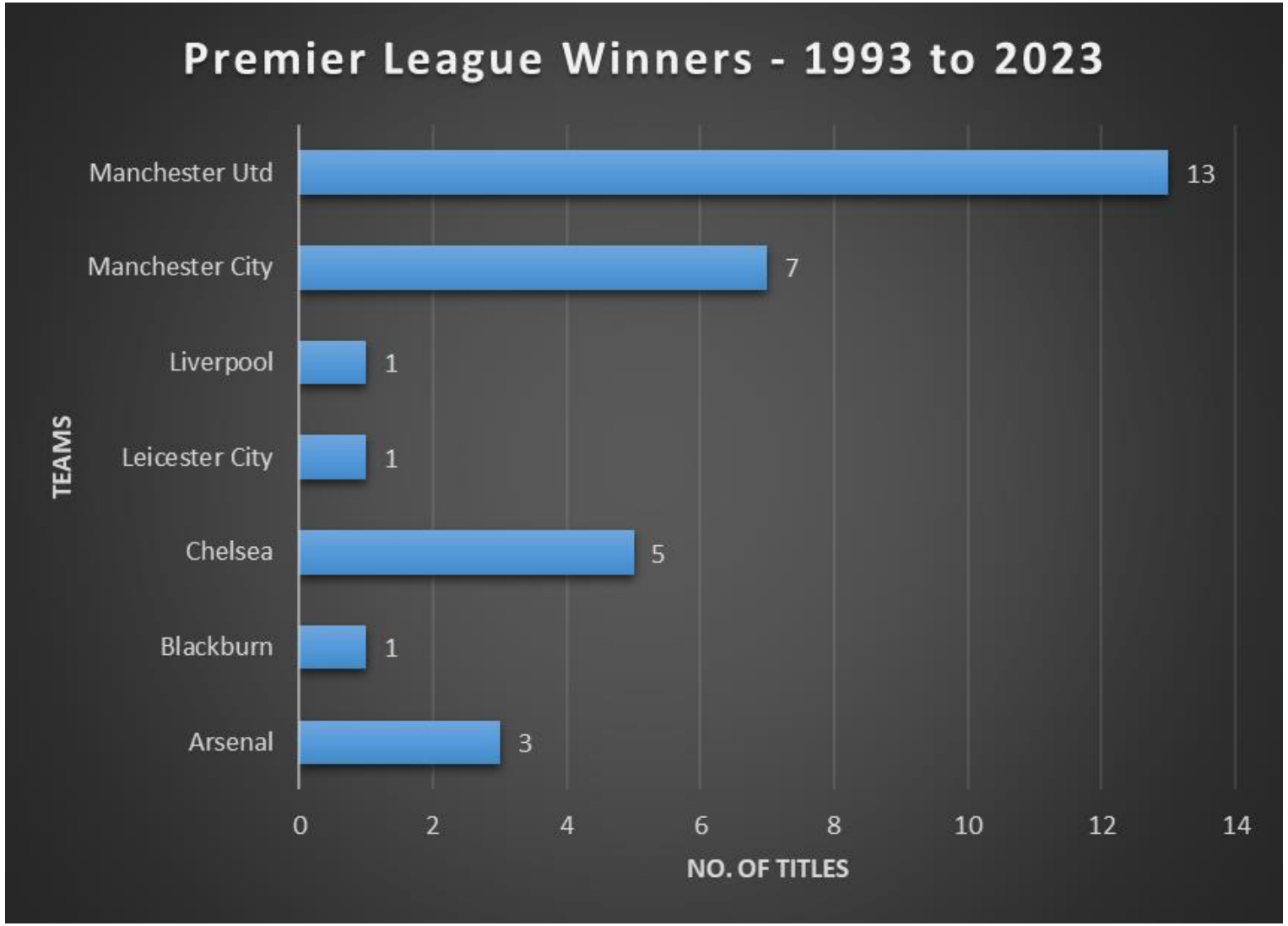
Points tables for each season, combined into one dataframe. Can access a particular season or seasons.

CONVERSION TO A POINTS TABLE

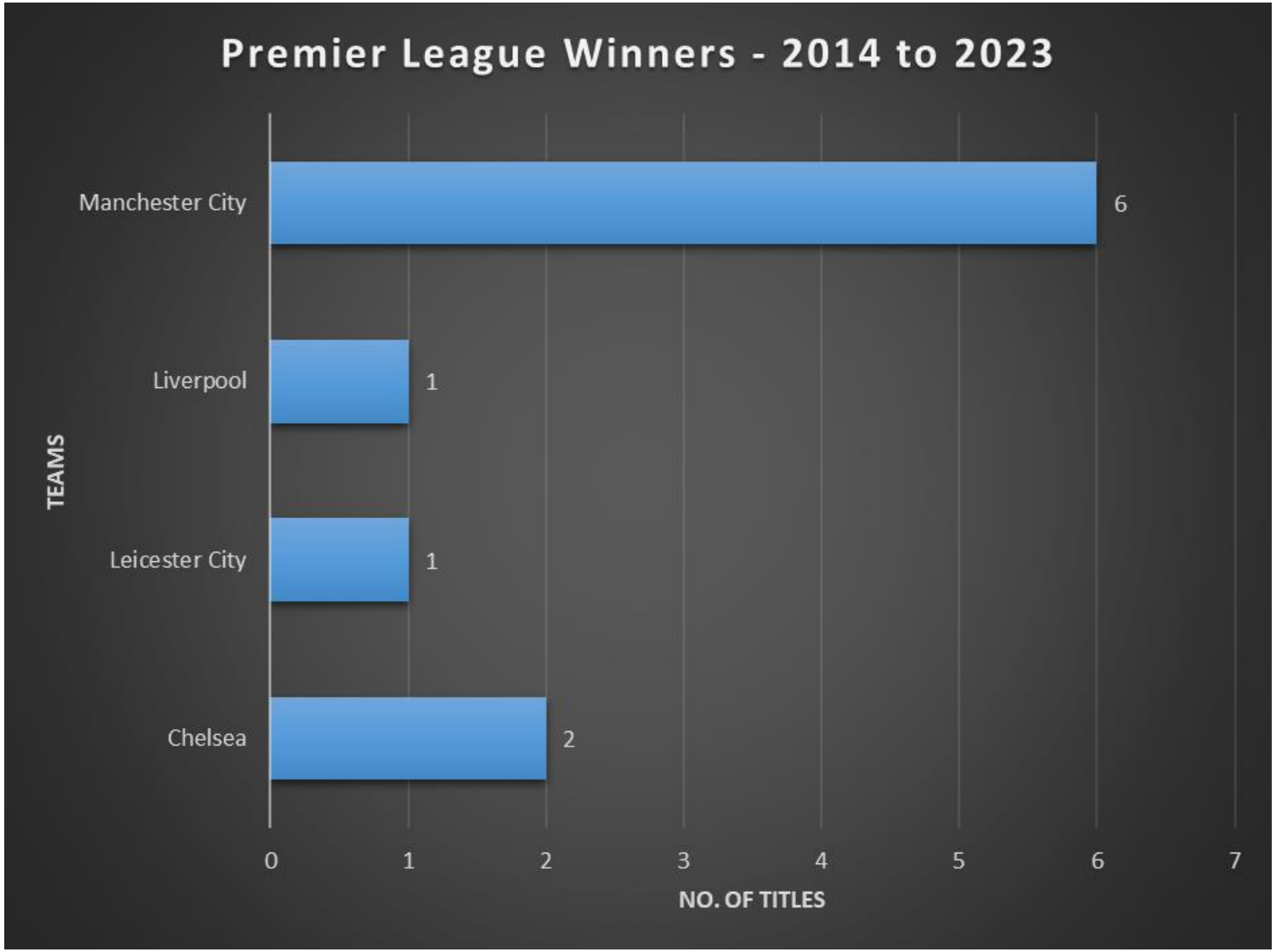
	Team	Played	Won	Drawn	Lost	GF	GA	GD	Points	Season_End_Year	Position
0	Manchester Utd	42	24	12	6	67	31	36	84	1993	1
1	Aston Villa	42	21	11	10	57	40	17	74	1993	2
2	Norwich City	42	21	9	12	61	65	-4	72	1993	3
3	Blackburn	42	20	11	11	68	46	22	71	1993	4
4	QPR	42	17	12	13	63	55	8	63	1993	5
...	...	...	...	...	...	...	...	...	...	...	...
621	Nott'ham Forest	38	9	11	18	38	68	-30	38	2023	16
622	Everton	38	8	12	18	34	57	-23	36	2023	17
623	Leicester City	38	9	7	22	51	68	-17	34	2023	18
624	Leeds United	38	7	10	21	48	78	-30	31	2023	19
625	Southampton	38	6	7	25	36	73	-37	25	2023	20

The complete points table, which is exported to Excel.

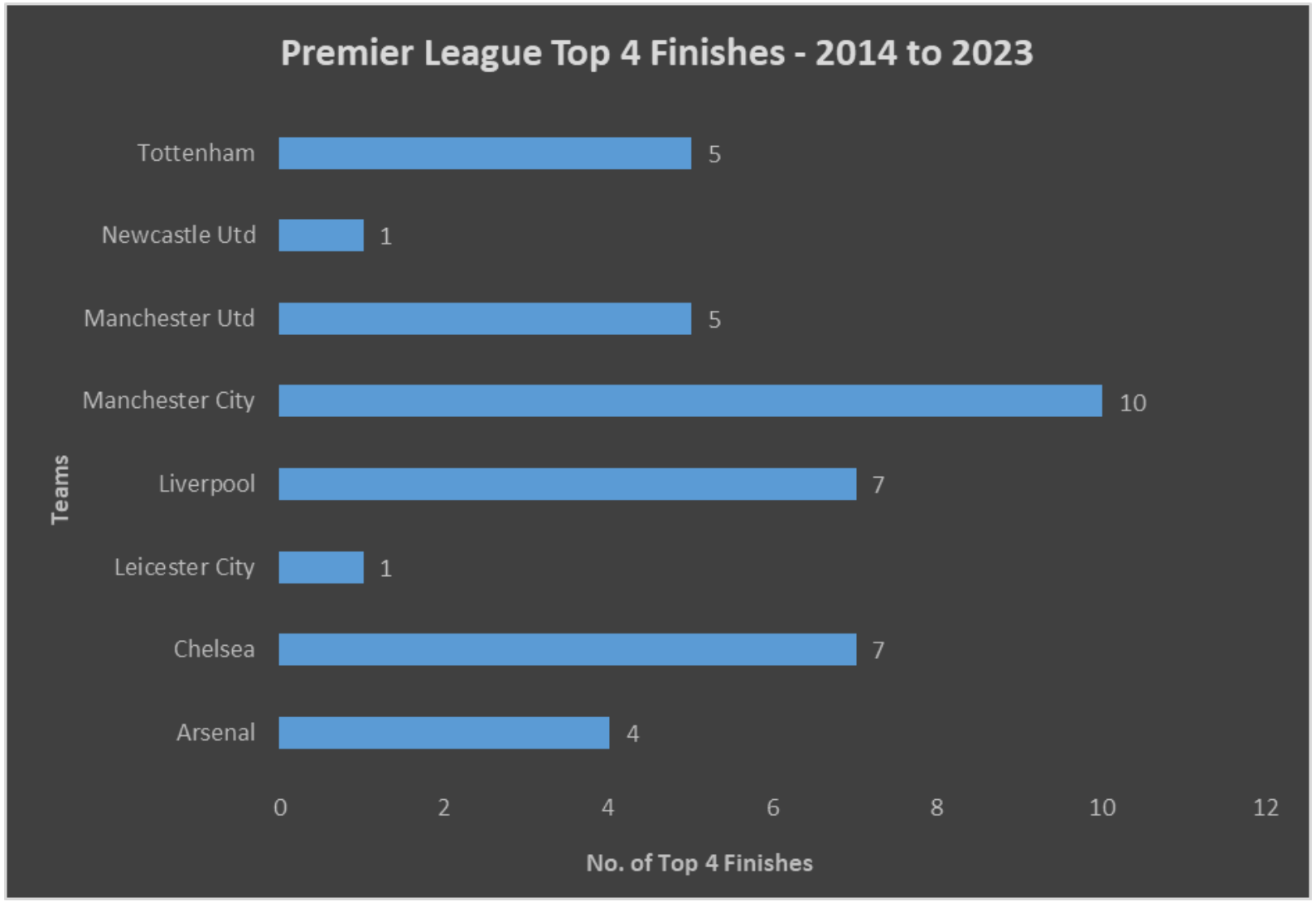
PREMIER LEAGUE WINNERS



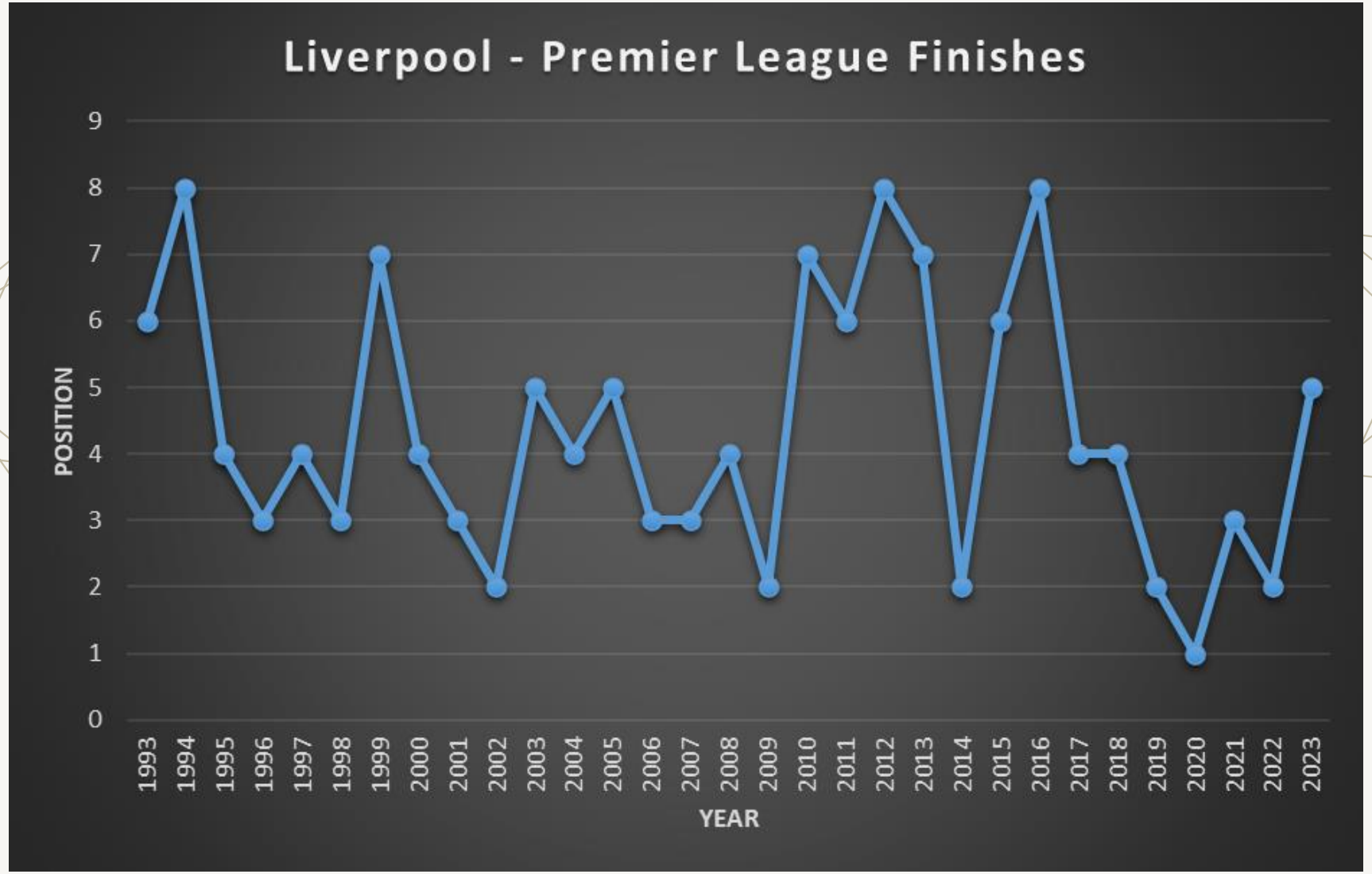
PREMIER LEAGUE WINNERS – LAST 10 YEARS



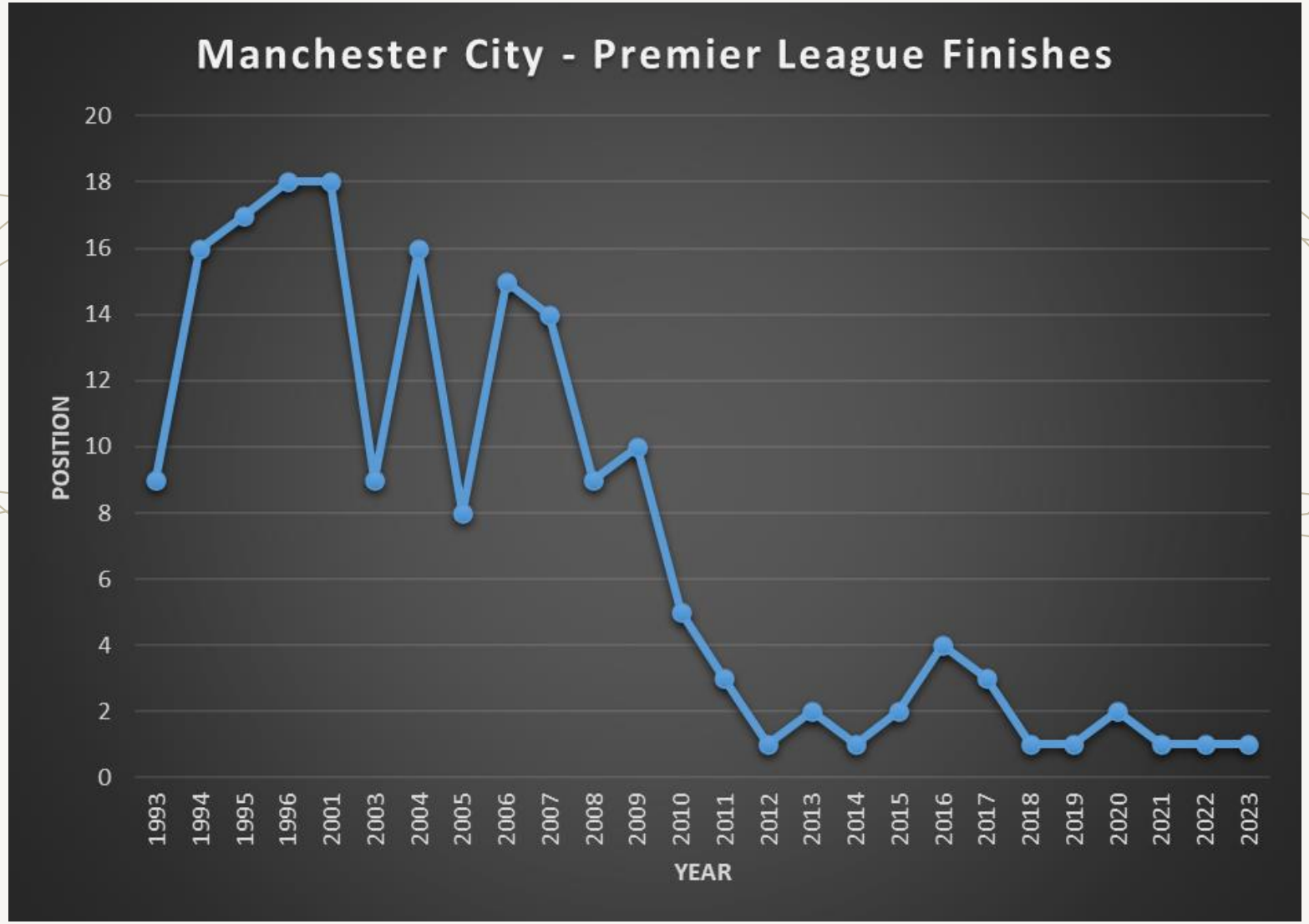
PREMIER LEAGUE TOP 4 – LAST 10 YEARS



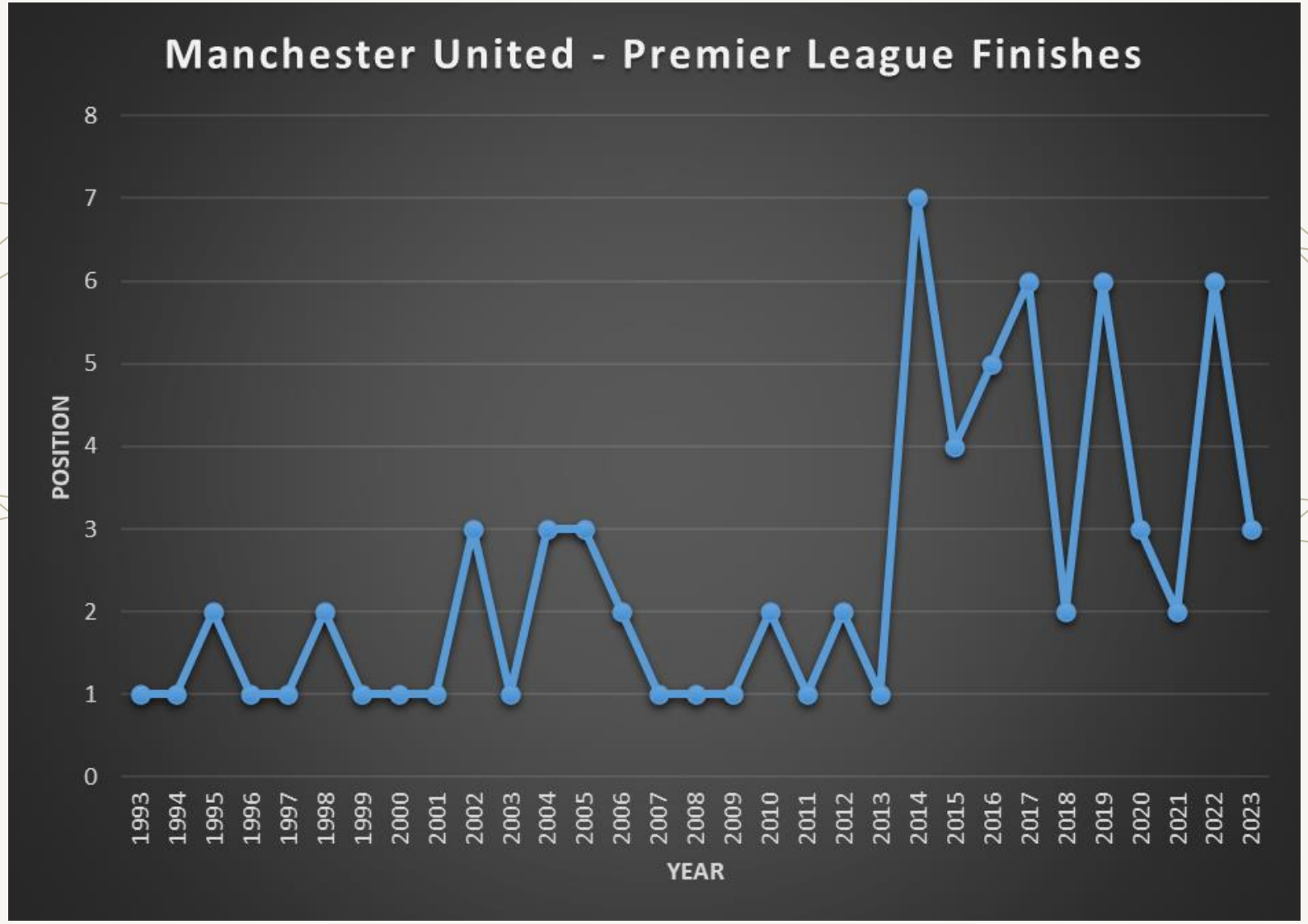
PERFORMANCE OF INDIVIDUAL TEAMS – LIVERPOOL



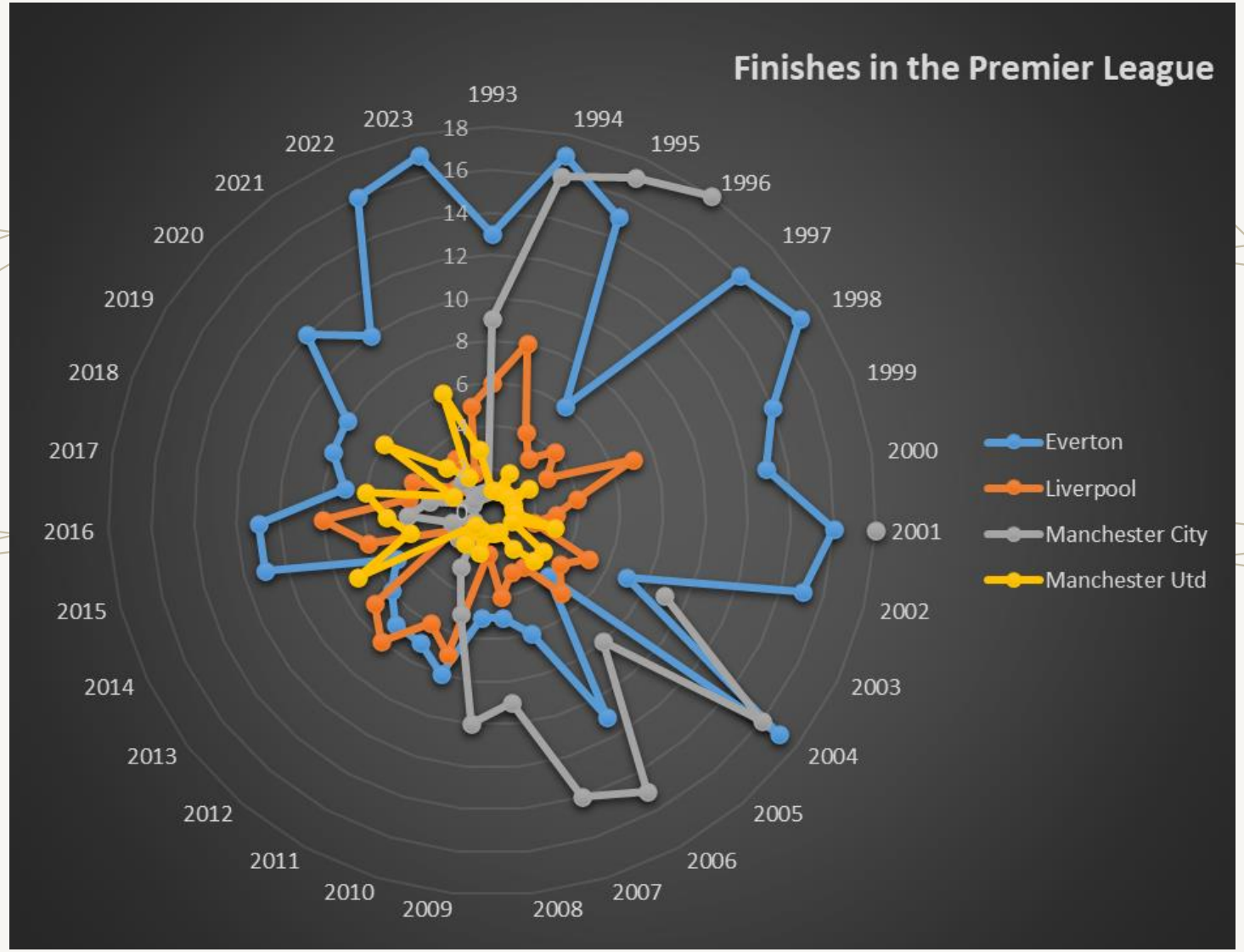
PERFORMANCE OF INDIVIDUAL TEAMS – MANCHESTER CITY



PERFORMANCE OF INDIVIDUAL TEAMS –MANCHESTER UNITED



ANOTHER WAY TO VISUALISE...

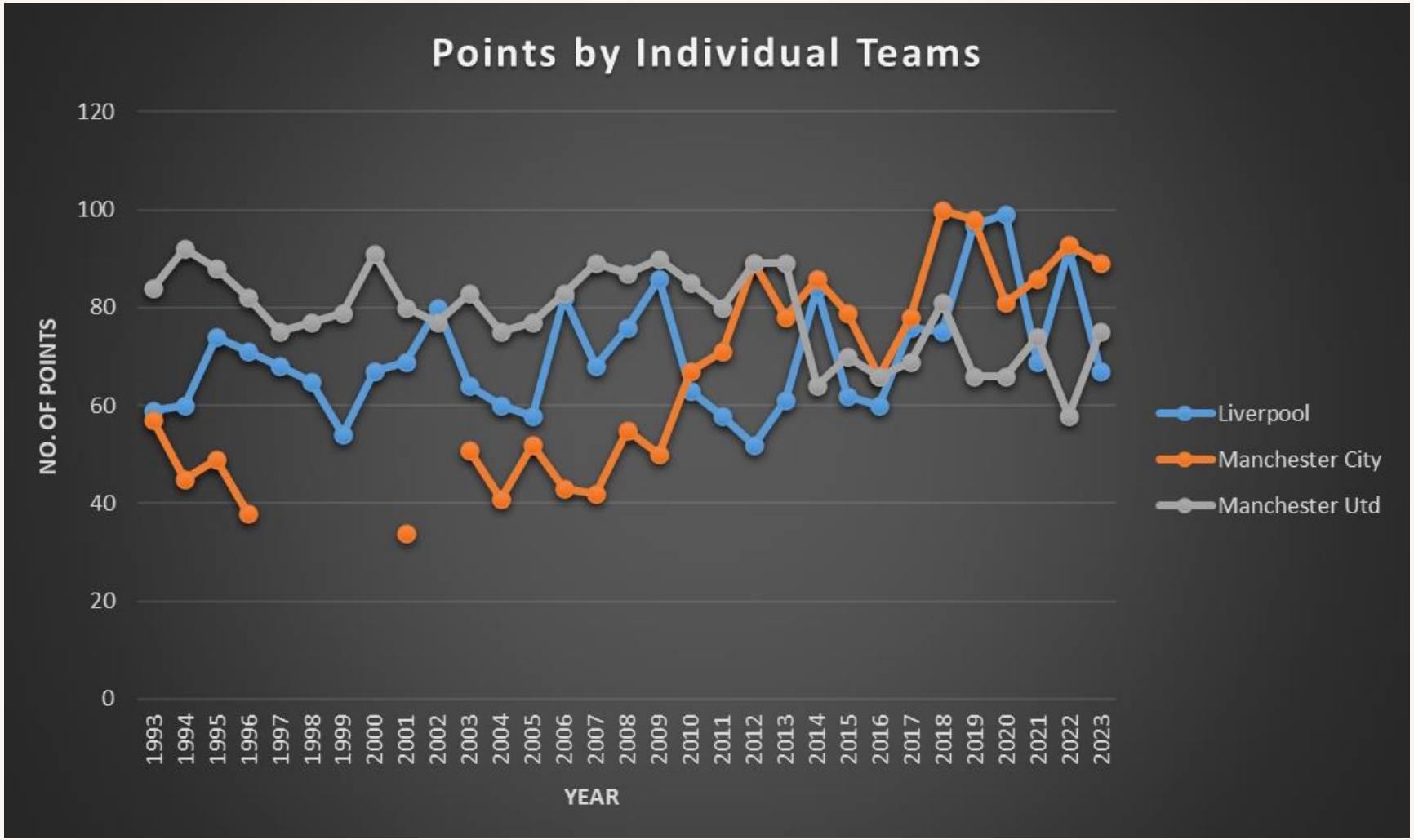


POINTS BY INDIVIDUAL TEAMS

Before – around 90+ points guaranteed the title

Now, even getting close to a 100 points might not be enough

The quality, consistency, and competitiveness has increased



WHAT ELSE?

Study the other sections of the table – mid-table, bottom half…

Relegation battles, how many points a team requires on average to avoid being relegated…

Discover the mid-table teams – those that never win the Premier League, but haven't been relegated either

So much data, so many things to visualise and analyse – we just need to ask the right questions

THANK YOU!

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